SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Duration: 1 Hour

QUESTIONS: 1 Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Answer:

Introduction

Artificial Neural Networks (ANNs) are widely used in educational data analysis to predict student performance based on features such as attendance, previous academic records, assignment scores, study hours, and examination marks. During model development, it is essential to evaluate how well the neural network performs on both the training dataset and unseen testing dataset.

One of the most commonly used performance metrics for regression models is **Mean Squared Error (MSE)**. MSE measures the average squared difference between the actual values and the predicted values. A lower MSE indicates that the predicted values are very close to the actual values, whereas a higher MSE indicates larger prediction errors.

The mathematical formula for MSE is:

$$\boxed{\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Where:

- y_i= Actual student score
- $\hat{y}_i$= Predicted student score
- n= Total number of observations

For the given neural network:

- **Training MSE = 0.85**

- **Testing MSE = 5.20**

These values indicate that the model performs very well during training but performs significantly worse during testing.

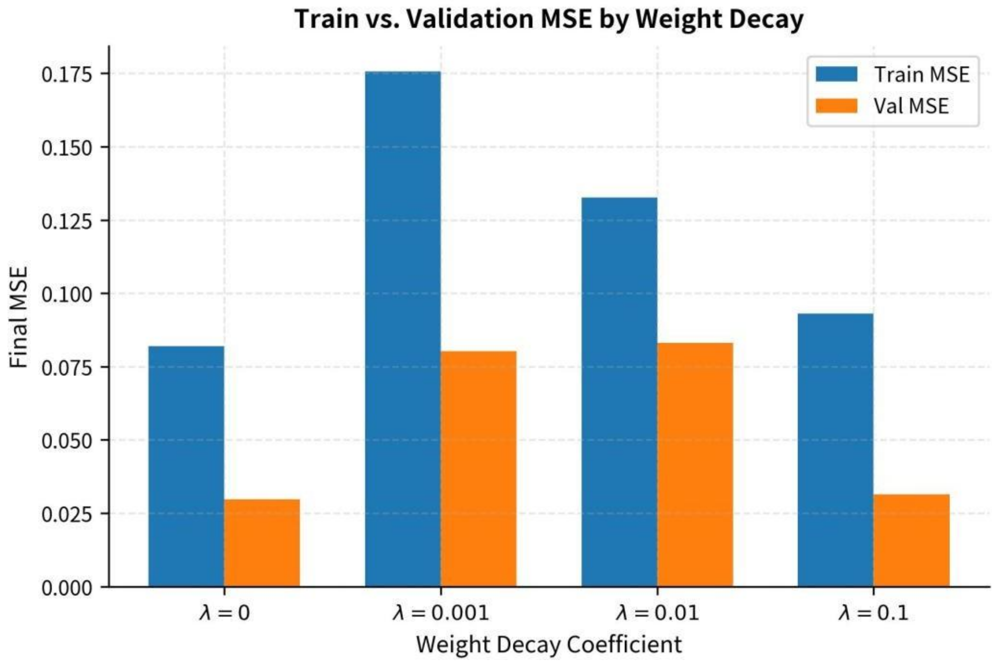

Figure 1: Comparison of Training and Testing Mean Squared Error (MSE).

Analysis

The neural network has achieved a training MSE of 0.85, which means that it has learned the relationships between the input features and student scores very effectively. The prediction errors on the training dataset are very small, indicating high accuracy during training. However, the testing MSE increases to 5.20, which is considerably larger than the training error. This means that when the model is applied to new student records that were not seen during training, its predictions become much less accurate.

The difference between the two errors is:

$$5.20 - 0.85 = 4.35$$

The testing MSE is approximately:

$$\frac{5.20}{0.85} \approx 6.1$$

This means the testing error is about six times larger than the training error. Such a large difference clearly indicates that the model has poor generalization capability.Instead of learning the underlying relationship between the input features and student performance, the neural network has memorized the training dataset. As a result, it performs well only on the data used for training but fails when predicting new student scores. This phenomenon is known as overfitting.

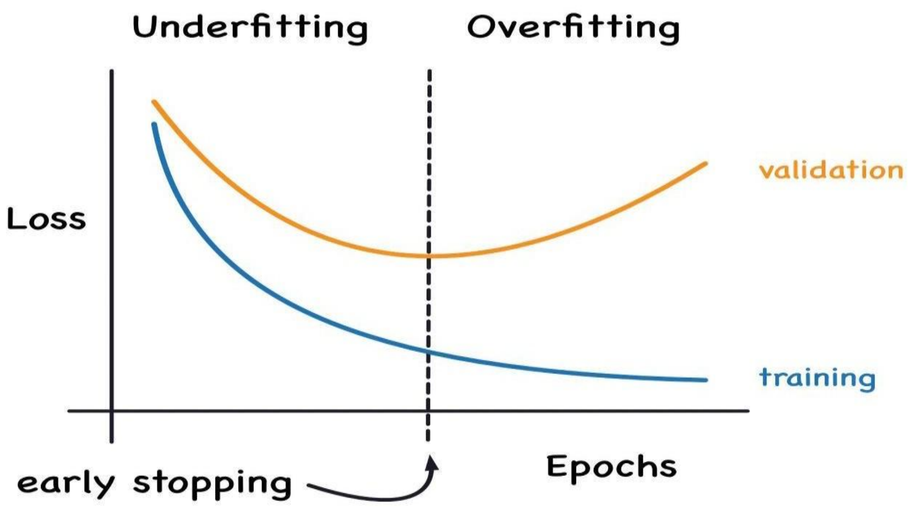

Figure 2: During overfitting, the training loss continues to decrease while the validation/testing loss begins to increase after a certain point.

Possible Reasons for the Difference

1. Overfitting
The neural network has memorized the training examples instead of learning generalized patterns. Consequently, it achieves excellent performance on training data but poor performance on unseen data.

2. Insufficient Training Data
If only a small number of student records are available, the model cannot learn all possible variations in student performance, reducing its prediction accuracy on new data.

3. High Model Complexity
A neural network with too many hidden layers, neurons, or trainable parameters may become unnecessarily complex and capture noise instead of meaningful patterns.

4. Presence of Noisy Data
Incorrect entries, missing values, outliers, or inconsistent records in the dataset can confuse the learning process and reduce prediction accuracy.

5. Difference Between Training and Testing Data
If the training and testing datasets follow different statistical distributions, the model may not be able to generalize effectively.

6. Lack of Regularization Techniques
Without methods such as Dropout or L1/L2 regularization, the model is more likely to memorize the training dataset.

7. Excessive Number of Training Epochs
Training the model for too many epochs may cause it to fit the training data too closely, reducing its ability to predict unseen data accurately.

Corrective Measures

To improve the performance of the neural network and reduce the testing error, the following corrective measures can be adopted:

- Increase the size and diversity of the training dataset.
- Apply L1 or L2 Regularization to prevent excessively large model weights.
- Introduce Dropout Layers to randomly deactivate neurons during training and improve generalization.
- Use Early Stopping to stop training when the validation error begins to increase.
- Simplify the neural network architecture by reducing unnecessary hidden layers or neurons.
- Perform K-Fold Cross Validation to obtain a more reliable estimate of model performance.
- Normalize and preprocess the data by handling missing values and removing outliers.
- Tune important hyperparameters such as the learning rate, batch size, number of epochs, and optimizer to achieve better performance.

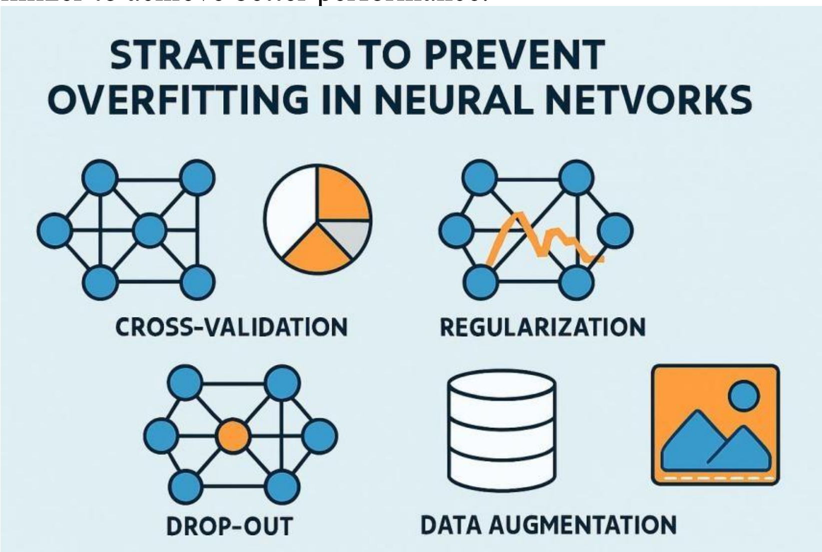

Figure 3: Common techniques used to reduce overfitting and improve model generalization.

Interpretation

The neural network demonstrates excellent performance on the training dataset, but its prediction accuracy decreases significantly on the testing dataset. The large gap between the two MSE values indicates that the model has not generalized well to unseen student data.

A well-trained neural network should have training and testing MSE values that are relatively close. The observed gap highlights the need for improved model design, better data preprocessing, and appropriate regularization techniques.

Conclusion

The neural network produced a training MSE of 0.85 and a testing MSE of 5.20. The substantial increase in testing error indicates that the model suffers from overfitting, resulting in poor generalization to new student data. This issue can be effectively addressed by increasing the amount of training data, applying regularization techniques, using dropout and early stopping, simplifying the model architecture, tuning hyperparameters, and performing proper data preprocessing. Implementing these measures will improve the model's prediction accuracy and overall reliability.

Code of Conduct:

I V Yogeswar(192425175)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Yogeswar

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary (5)**	**Level 3 – Proficient (4)**	**Level 2 – Developing (2-3)**	**Level 1 – Beginning (0-1)**
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided